

Ngoc Thanh Nguyen · Vu Dinh Duc Anh ·
Adrianna Kozierekiewicz ·
Sinh Nguyen Van · Manuel Nunez ·
Jan Treur · Gottfried Vossen (Eds.)

Communications in Computer and Information Science 2748

Advances in Computational Collective Intelligence

17th International Conference, ICCCI 2025
Ho Chi Minh City, Vietnam, November 12–15, 2025
Proceedings, Part II

Part 2

Ngoc Thanh Nguyen · Vu Dinh Duc Anh ·
Adrianna Koziarkiewicz · Sinh Nguyen Van ·
Manuel Nunez · Jan Treur · Gottfried Vossen
Editors

Advances in Computational Collective Intelligence

17th International Conference, ICCCI 2025
Ho Chi Minh City, Vietnam, November 12–15, 2025
Proceedings, Part II

Editors

Ngoc Thanh Nguyen 
Wrocław University of Science
and Technology
Wrocław, Poland

Adrianna Kozierekiewicz 
Wrocław University of Science
and Technology
Wrocław, Poland

Manuel Nunez 
Universidad Complutense de Madrid
Madrid, Spain

Gottfried Vossen 
University of Münster
Münster, Germany

Vu Dinh Duc Anh 
VNU-HCM International University
Ho Chi Minh City, Vietnam

Sinh Nguyen Van 
VNU-HCM International University
Ho Chi Minh City, Vietnam

Jan Treur 
Vrije Universiteit Amsterdam
Amsterdam, The Netherlands

ISSN 1865-0929 ISSN 1865-0937 (electronic)
Communications in Computer and Information Science
ISBN 978-3-032-10208-9 ISBN 978-3-032-10209-6 (eBook)
<https://doi.org/10.1007/978-3-032-10209-6>

© The Editor(s) (if applicable) and The Author(s), under exclusive license
to Springer Nature Switzerland AG 2026

This work is subject to copyright. All rights are solely and exclusively licensed by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.

The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.

The publisher, the authors and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

If disposing of this product, please recycle the paper.

An Explainable AI-Based Breast Cancer Classification Combining Rule Sets and Deep Neural Networks	260
<i>Ho-Dat Tran, Thuong-Cang Phan, Vinh-Phong Nguyen, and Anh-Cang Phan</i>	

Data Mining and Machine Learning

Prediction of Real Estate Prices in 15 Largest Polish Cities Using Machine Learning	277
<i>Ewa Walaszczyk, Tomasz Paliński, Artur Rot, and Karol Łopaciński</i>	
Automated Stock Trading with Q-Learning Algorithm	290
<i>Dong Quan Pham and Thi Thanh Sang Nguyen</i>	
EpiPredict: A Smart Solution for Real-Time Epileptic Seizure Prediction and Alerts Based on Machine Learning Models	304
<i>Rehab Nailly and Mourad Zaied</i>	
Using Closed Frequent Itemsets To Explain Classification Tasks	323
<i>Thao Ngo and Bac Le</i>	
Measuring Causal Effects of Online Medical Records on Medical Decision Making Using Causal Machine Learning	337
<i>Minh Nguyen and Long Hoang Le</i>	
Machine Learning-Based Chat Bot for Diagnosing Depression: A Mental Disorder Illness	352
<i>Mohamed Torky, Najib Ben Aoun, Sadique Ahmad, Menna Lashin, Sohaila Abobakr, Mohamed El-Mogy, and Aboul Ella Hassanein</i>	
LogiDRS: Logical Reasoning Using Discourse Relation Sense	364
<i>Toan Pham, Tien Le, Hoang Quoc Vu, and Nhi-Thao Tran</i>	
From Networks to Circuits: A Structured Approach to AI Interpretability	380
<i>Bianka Kowalska and Halina Kwaśnicka</i>	

Cybersecurity, Blockchain Technology and Internet of Things

IVFDT: An Improved Very Fast Decision Tree Approach for Blockchain Data Stream Classification	397
<i>Ahmed Faris Alsayyad and Mohamed Mabrouk</i>	



An Explainable AI-Based Breast Cancer Classification Combining Rule Sets and Deep Neural Networks

Ho-Dat Tran^{1,2} , Thuong-Cang Phan¹ , Vinh-Phong Nguyen³ ,
and Anh-Cang Phan²

¹ Can Tho University, CanTho, Vietnam

`ptcang@cit.ctu.edu.vn`

² Vinh Long University of Technology Education, VinhLong, Vietnam

`{datth,cangpa}@vlute.edu.vn`

³ Can Tho Oncology Hospital, CanTho, Vietnam

Abstract. Cancer has always garnered significant attention due to its prevalence and the impact it has on many individuals. Recent advancements in artificial intelligence (AI) have led to numerous research initiatives that effectively assist doctors in diagnosis and treatment, thus enhancing patients' chances of recovery. Mammography remains one of the most commonly used screening methods today. In this paper, we propose a detection and classification method for mammography that integrates a fuzzy inference system (FIS) with explainable artificial intelligence (XAI) to enhance classification efficiency and improve the interpretation of outputs from machine learning models. We conduct experiments across various scenarios to assess the model's practical applicability. The experimental results demonstrate an accuracy exceeding 94.64% and an IoU index of 0.899, confirming the proposed method's effectiveness. These findings lay a crucial foundation for our ongoing development of fuzzy learning techniques aimed at tumor classification and identification in mammography.

Keywords: Breast cancer · FIS · XAI · Deep learning

1 Overview

Breast cancer remains the leading cause of cancer-related mortality among women in both developing and developed nations. The World Health Organization (WHO) estimates that by 2025 there will be approximately 19.3 million newly diagnosed cases worldwide [2]. As the most prevalent form of cancer in women, it affects around 2.1 million individuals each year and accounts for a substantial share of female cancer deaths. Timely detection and precise classification are crucial, as they allow patients to receive appropriate treatments, reduce the

H.-D. Tran and T.-C. Phan—Equal contribution.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
N. T. Nguyen et al. (Eds.): ICCCI 2025, CCIS 2748, pp. 260–274, 2026.
https://doi.org/10.1007/978-3-032-10209-6_18

necessity for invasive surgeries, and ultimately decrease mortality rates. Mammograms detect abnormalities such as masses, calcifications, architectural distortions, and asymmetries. Since 2D mammograms involve image superposition, multiple breast views may be necessary for comprehensive information. Standard mammography includes two views for each breast: the craniocaudal (CC) view from above and the mediolateral oblique (MLO) view from an angle. Additional views may be taken as needed. Digital breast tomosynthesis (DBT), or 3D mammography, now provides enhanced spatial context for detailed breast examination and is becoming the standard of care [1]. Clancy et al. [3] investigated the effectiveness of pre-training procedures in CNNs for classifying mammography pictures as benign, malignant, or recalled-benign. Six pre-training procedures were examined, including transfer learning with medical and non-medical datasets, layer freezing, and alternative network topologies (AlexNet, ResNet-152). The dataset included 4935 mammography pictures from 1303 patients and four pre-training datasets. AlexNet pre-trained on ImageNet and Breast Density has the highest accuracy (AUC) of 0.68–0.77. Apoorva et al. [4] suggested a method utilizing CNNs for image analysis and machine learning algorithms like KNN, CART, SVM, and Naïve Bayes for numerical datasets, using breast tumor image features. The key contribution is a completely automated breast cancer categorization and prediction system that uses a Flask-based GUI to connect the model to input data and forecast benign or malignant breast cancer. CNN achieved 98.13% accuracy on IDC set, SVM 96.48% on numerical set. Lizzi et al. [12] study on AI (XAI) in breast density diagnosis from mammograms used two major methodologies. After training and testing a 41-layer Residual CNN model on 1800 mammograms (7200 pictures). Additionally, Grad-CAM was utilized to explain model decisions. Prodan et al. [5] used StyleGAN-XL to generate synthetic images from mammography images and data augmentation approaches to classify breast cancer. The authors improved classification performance on imbalanced datasets with synthetic images and employed explainable AI approaches like class activation maps and bounding boxes to comprehend model decision-making. Synthetic augmentation yielded the highest accuracy with 512×512 images: ResNet18 0.85, ResNet34 0.84, EfficientNetB0 0.84, and MaxViT 0.88, with EfficientNetB0 having the highest F1 score of 0.51.

2 Background

2.1 U-Net

U-Net, developed by Ronneberger et al. [10], is a neural network designed for image segmentation. The network has a symmetric structure with two main paths: contraction and expansion. The contraction path (encoder) on the left side functions like a standard convolution network to provide classification information. The expansion path (decoder) on the right side uses deconvolution layers for learning and segmentation. Short connections between matching levels prevent information loss during the convolution and deconvolution processes. In the

contraction path, the network extracts and learns image features while reducing image size through convolutions and pooling layers to create high-resolution feature maps. The expansion path enables the network to learn detailed classification information for specific areas. information.

2.2 DeepLabV3+

DeepLabv3+ [11] is an advanced deep learning model designed for image segmentation, especially semantic segmentation. It is an improved version of DeepLabv3+, adding a decoder module to improve the granularity of segmentation results, especially at object boundaries. DeepLabv3+ uses Atrous Convolution and Atrous Spatial Pyramid Pooling (ASPP) neural networks to expand the perception range without losing resolution, thereby improving the ability to capture contextual information at different dimensions. DeepLabv3+ stands out for its ability to balance high accuracy and computational efficiency, making this network model an ideal choice for real-time applications or on resource-constrained devices. In addition, this model can be easily integrated with popular backbone architectures such as Xception or ResNet to optimize performance.

2.3 Explainable Artificial Intelligence (XAI)

AI applications and research have advanced due to Deep Neural Networks (DNNs). Though successful, most machine learning algorithms cannot explain their outputs, leaving consumers to wonder why the neural network reached a conclusion. Thus, AI model interpretation is essential. XAI provides tools and methods for analyzing and explaining deep learning model outputs. It lets people engage in AI decision-making loops and interrupt or regulate its tasks. AI systems must do jobs well and provide transparent reports on why and how they did them. Popular XAI Methods: Global Model Interpretability: Overall model explanation is the goal of this method. Simple models like decision trees and linear regression are used, making them straightforward to understand. Local Model Interpretability: This method explains individual predictions rather than the complete model. LIME and SHAP are common approaches. Charts or visual displays (like heatmaps in image recognition algorithms) to show which data factors affect results.

In this paper, we implement visualization techniques with Grad-Cam heatmaps, which is a method for explaining the results of deep learning models in the field of image processing. Grad-CAM generates a heatmap based on the most important features the model uses to make its prediction. This heatmap visualizes how much attention the model pays to each region of the image, which is intended to better explain the extent of the mass produced on the mammogram.

2.4 Fuzzy Logic

Fuzzy logic, as described by Zadeh [9], is a method that is utilized for the purpose of processing information and making judgments when there are values that

are ambiguous or uncertain. In contrast to classical logic, which only works with binary values (zero or one), fuzzy logic allows for a range of values between zero and one. This wide range of values makes it possible to describe states that are neither completely true nor completely untrue. The term “fuzzy machine learning” refers to the process of incorporating fuzzy learning approaches into the field of machine learning. These models are very useful for managing unclear information that is frequently encountered in medical imaging. They also improve the inference and analysis of aberrant regions within pictures, which ultimately leads to an improvement in the effectiveness of diagnosis and categorization. The application of fuzzy rules and a broad knowledge base are two of the ways in which a fuzzy inference system makes decision-making easier.[par] Fuzzification, fuzzy inference, and defuzzification are the three stages that make up the fuzzy logic process. These stages are responsible for handling input and output variables. During the process of fuzzification, crisp values are transformed into fuzzy values by mapping actual inputs to membership functions and computing the membership grade of each input. The crisp values of the input are converted into fuzzy values with the assistance of fuzzy membership functions. These fuzzy values describe the degree to which an element belongs to a fuzzy set for the input. In the context of a fuzzy set M , the membership function is defined by the expression $\theta_M(x)$, where $\theta_M(x) \in [0, 1]$ denotes the degree of membership of x to the fuzzy set M . A graphical representation of each membership function is presented in order to highlight the degree to which each input contributes to a set. A mathematically straightforward approach is provided by the triangular membership function, which is utilized in this study. It is defined by three parameters $\{a, b, c\}$, where the membership function $\theta_M(x)$ for each value x is specified in Eq. 1. The values of these parameters are taken from the knowledge of specialists.

$$\theta_M(x) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x-a}{b-a} & \text{if } a < x \leq b \\ \frac{c-x}{c-b} & \text{if } b < x \leq c \\ 0 & \text{if } x > c \end{cases} \quad (1)$$

A fuzzy inference is performed during the second phase, during which the system processes the values that are input. Concluding statements are generated by the fuzzy inference system by employing IF-THEN language rules derived from a knowledge base. For the purpose of influencing the fuzzy output sets and the conclusions that are ultimately drawn, these rules use input membership values as weights. During the last phase, the fuzzy findings are converted back into exact and crisp values by the process of defuzzification. The method of defuzzification involves taking a weighted average of the values of x , where the weights are the membership function values denoted by $\theta(x)$.

$$\odot = \frac{\int x \cdot \theta(x) dx}{\int \theta(x) dx} \quad (2)$$

where $\theta(x)$ is the membership function (fuzzy attribute level) of the value x ; x is a continuous input variable.

2.5 Rule-Based Algorithm

- **JRIP**: is an implementation of the RIPPER (Repeated Incremental Pruning to Produce Error Reduction) algorithm [7], a rule-based classification algorithm known for its simplicity. It generates efficient, easy-to-understand rule sets through an iterative process. These rules are refined through pruning to minimize test set errors. JRIP applies rules in priority order, with the first matching rule taking precedence.

Algorithm 1. JRIP Alg.

Require: Training dataset $\mathcal{D}$ with categories $\{C_1, C_2, \dots, C_m\}$

Ensure: Learned rule collection $\mathcal{R}$

- 1: Order the categories $\{C_1, C_2, \dots, C_m\}$ in ascending order of sample counts.
 - 2: Initialize $\mathcal{R} \leftarrow \emptyset$.
 - 3: **for** each category C_j in the ordered list **do**
 - 4: **repeat**
 - 5: Construct a provisional rule ρ_j that maximally fits the samples of C_j .
 - 6: Refine ρ_j through pruning and optimization steps.
 - 7: Insert ρ_j into $\mathcal{R}$.
 - 8: Eliminate all instances of $\mathcal{D}$ that are covered by ρ_j .
 - 9: **until** no instances of C_j remain uncovered
 - 10: **end for**
 - 11: Define a default rule to assign labels for any instances not captured by $\mathcal{R}$.
 - 12: **return** $\mathcal{R}$ as the final induced rule base.
-

- **PART**: combines decision trees [8] and rule-based methods into a hybrid algorithm. Instead of building complete decision trees, PART creates rules from partial tree segments. It works by constructing a partial decision tree, extracting a rule from one of its branches, removing the cases covered by that rule, and repeating this process until no cases remain.

Algorithm 2. PART Alg.

Require: Training dataset $\mathcal{D}$

Ensure: Derived rule base $\mathcal{R}$

- 1: Start with an empty rule collection: $\mathcal{R} \leftarrow \emptyset$
 - 2: Define $\mathcal{D}_{\text{uncovered}} \leftarrow \mathcal{D}$ to represent instances not yet explained
 - 3: **while** $\mathcal{D}_{\text{uncovered}} \neq \emptyset$ **do**
 - 4: Induce a partial decision tree using $\mathcal{D}_{\text{uncovered}}$
 - 5: Select the most discriminative branch to form a provisional rule φ
 - 6: Append φ into $\mathcal{R}$
 - 7: Remove from $\mathcal{D}_{\text{uncovered}}$ all samples covered by φ
 - 8: Discard the remainder of the partial tree
 - 9: **end while**
 - 10: **return** $\mathcal{R}$ as the final constructed rule base
-

2.6 Evaluation Metrics

To evaluate the object detection model, we use Precision and Recall metrics. Precision measures prediction accuracy as the ratio of true positives (TP) to total predicted cases, including false positives (FP), calculated using Eq. 3. Recall measures sensitivity as the ratio of true positives to total positive cases, including false negatives (FN), calculated using Eq. 4.

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

$$F1_score = 2 \frac{Precision * Recall}{Precision + Recall} \quad (5)$$

In segmentation frameworks such as U-Net and DeepLabv3+, the loss function is generally formulated to quantify the gap between the predicted segmentation map and the ground truth labels. In our study, both loss and accuracy are evaluated using the following mathematical expressions:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

To calculate the average precision over all classes we use the mAP [6] measure given by the formula:

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (7)$$

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union} \quad (8)$$

$$Dice\ Coefficient = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (9)$$

$$Dice\ Loss = 1 - Dice\ Coefficient \quad (10)$$

where, True Positives (TP): the number of positive class (foreground) samples or pixels predicted by the model; TN (True Negatives): the number of negative class (or background) samples or pixels the model correctly predicts; FP (False Positives): the number of samples or pixels that belong to the negative class (or background) but are incorrectly predicted as the positive class (or foreground); FN (False Negatives): the number of samples or pixels that are positive (foreground) but predicted as negative by the model.

3 Proposed Method

This study introduces a unified four phases, which consists of: (1) preprocessing to prepare the input data for deep learning, (2) extracting features and forming rule sets, (3) training the classification model based on these rule sets, and (4) interpreting results with XAI. The overall architecture of the proposed approach is depicted in Fig. 1.

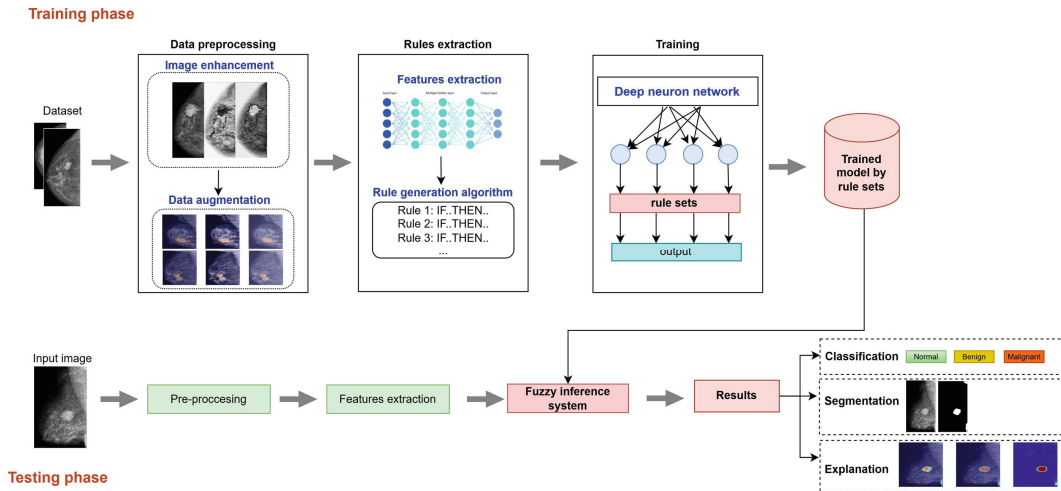


Fig. 1. The proposed model for breast cancer segmentation and classification.

- **Data Pre-processing:** Each input image is resized to 224×224 pixels, saved in .jpg format, and normalized prior to training. Pre-processing also involves contrast enhancement using histogram-based techniques. Furthermore, data augmentation strategies are applied to increase the diversity of training samples.

- **Feature Extraction and Rule Generation:** After pre-processing, the images are fed into deep learning models to extract relevant features. These features are then transformed into rule sets through the application of the JRIP and PART algorithms, as described in Sect. 2.5.

- **Classification and Interpretation with FIS:** The classification stage employs a Fuzzy Inference System (FIS). The system is trained using features obtained from U-Net and DeepLabv3+ models, which are converted into fuzzy rules. These rules associate fuzzy inputs with their corresponding class labels and define the membership functions for each feature. During inference, features from unseen images are fuzzified and processed through the FIS to generate fuzzy outputs. A defuzzification step is then performed using the weighted average method, producing crisp classification results. Finally, these results are matched with the trained classes, enabling accurate categorization of the test images. This pipeline ensures robust classification while retaining interpretability through fuzzy reasoning.

- **Explanation of Grad-CAM Heatmap:** The Grad-CAM heatmap visualizes the model's attention regions when making predictions. The heatmap intensity ranges from 0.0 (Blue regions) to 1.0 (Red regions). Intermediate values from 0.0 to 1.0 (shades of green to yellow) represent varying levels of attention or suspicion for potential abnormalities. The color gradient helps identify areas of interest, assisting physicians in further examination (Table 1).

Algorithm 3. Classification using Fuzzy Inference System (FIS) with Grad-CAM Integration**Input:** Trained model T , feature vector of the test sample X_{test} , FIS **Output:** Predicted label C

- 1: Standardize X_{test} so that all attributes are comparable in scale.
- 2: Create an empty container Fuzzy_values to store membership responses.
- 3: **for** each component x_i in X_{test} **do**
- 4: **for** each fuzzy set A_k defined in FIS **do**
- 5: Evaluate the degree of membership $\theta_{M_k}(x_i)$ via the membership function (Eq. 1).
- 6: Store $\theta_{M_k}(x_i)$ into Fuzzy_values.
- 7: **end for**
- 8: **end for**
- 9: Initialize Rule_activations as an empty list.
- 10: **for** every rule R encoded in FIS **do**
- 11: Derive the firing strength of R using Fuzzy_values with fuzzy operators.
- 12: Record this strength into Rule_activations.
- 13: **end for**
- 14: Combine the elements in Rule_activations to obtain crisp decision scores S_c for all classes.
- 15: Translate S_c into final numerical outputs using $\odot$ defuzzification method (Eq. 2).
- 16: Execute Grad-CAM on T to generate heatmaps highlighting regions that influenced the prediction.
- 17: Assign the output label as $C \leftarrow \arg \max S_c$.
- 18: **return** C

Table 1. Rule sets generated by the trained models

Algorithm	Rule sets
U-Net_JRIP	Rule 1: If fea_170 ≤ -0 AND fea_42 ≥ 0.000015 AND fea_173 ≥ -0 AND fea_31 ≥ 0.000002 , then class = 1 ... Rule 15: If fea_155 > 0.000287 AND fea_71 > -0.000267 AND fea_108 ≤ 0.000355 , then class = 0
U-Net_PART	Rule 1: If fea_133 ≤ -0.000521 AND fea_39 ≤ 0.001293 AND fea_178 > -0.000067 , then class = 1 ... Rule 196: If fea_64 ≤ 0.000749 , then class = 1
DeepLabv3+_JRIP	Rule 1: If fea_14 ≤ -0.051902 AND fea_16 ≤ -0.015505 AND fea_7 ≥ 0.06806 AND fea_21 ≥ -0.048258 AND fea_22 ≥ -0.084241 , then class=0 ... Rule 31: If fea_3 ≥ 0.434534 AND fea_28 ≥ 0.029928 AND fea_172 ≤ 0.001823 , then class=1
DeepLabv3+_PART	Rule 1: If fea_159 ≤ -0.012164 AND fea_87 > -0.009701 AND fea_138 ≤ 0.003037 , then class = 2 ... Rule 89: If fea_69 ≤ -0.000479 AND fea_182 > 0.000891 AND fea_158 > -0.006177 AND fea_161 ≤ -0.004041 AND fea_185 ≤ 0.001751 , then class = 0

4 Experimental Results

4.1 Evaluation Scenarios

The proposed method was assessed under six different experimental scenarios, each configured with the same training parameters (Table 2):

Table 2. Experimental scenarios along with their training parameters

Scenario	Name	Learning rate	Epoch	Batch size	Num classes	Num rules
1	U-Net	0.0001	100	32	3	-
2	U-Net_JRIP	0.0001	100	32	3	15
3	U-Net_PART	0.0001	100	32	3	196
4	DeepLabv3+	0.0001	100	32	3	-
5	DeepLabv3+_JRIP	0.0001	100	32	3	31
6	DeepLabv3+_PART	0.0001	100	32	3	89

In particular, Sce. 2, 3, 5, and 6 employ rule generation algorithms that derive rule sets from features extracted by deep learning models. This step aims to enhance classification performance by addressing ambiguous or conflicting patterns within the dataset. After preliminary trials with multiple parameter sets, we fixed identical hyperparameters across all scenarios: a learning rate of 0.0001, trained over 100 epochs, and three output classes representing benign, malignant, and normal cases.

4.2 Implementation Environment and Dataset

–**Implementation Environment:** All experiments were executed in a Python environment on Kaggle. The system configuration includes an Intel Xeon 2.00 GHz CPU with 30 GB RAM, and a Tesla P100 GPU with 16 GB RAM. TensorFlow v1.5 served as the core library for model training.

–**Dataset:** The experimental dataset consists of images collected from the VinDr repository combined with supplementary hospital data. In total, 242 benign, 310 malignant, and 493 normal mammograms were used. The dataset was split into training and testing sets at a ratio of 70:30.

4.3 Training Results

This subsection summarizes the performance evaluation of the proposed scenarios. Metrics considered include Loss, Accuracy, Precision, Recall, F1-score, IoU, as well as confusion matrices, following the definitions provided in Sect. 2.6. Training and testing durations were also recorded along with the classification and detection results.

The loss values obtained across the six scenarios were 0.0538, 0.0401, 0.0377, 0.1417, 0.0715, and 0.1191, respectively. Among these, Sce. 3 and 6 demonstrated more stable convergence compared to the others. However, when applying the JRIP algorithm (Sce. 2 and 5), the generated rule sets omitted certain key image features, which led to information loss and negatively influenced both training stability and classification accuracy. This finding highlights the crucial role of the rule generation process in determining the effectiveness of the overall model.

To further validate the robustness of the algorithms, we performed a consistency check on the dataset after applying JRIP and PART to the extracted rule sets. The results presented in Table 3 indicate that the JRIP-based approach achieved superior reliability in both the U-Net and DeepLabv3+ cases, providing an essential foundation for maintaining strong performance during testing (Figs. 2 and 3).

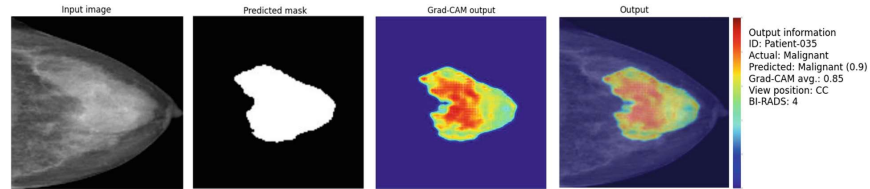


Fig. 2. Results on input image of the proposed model.

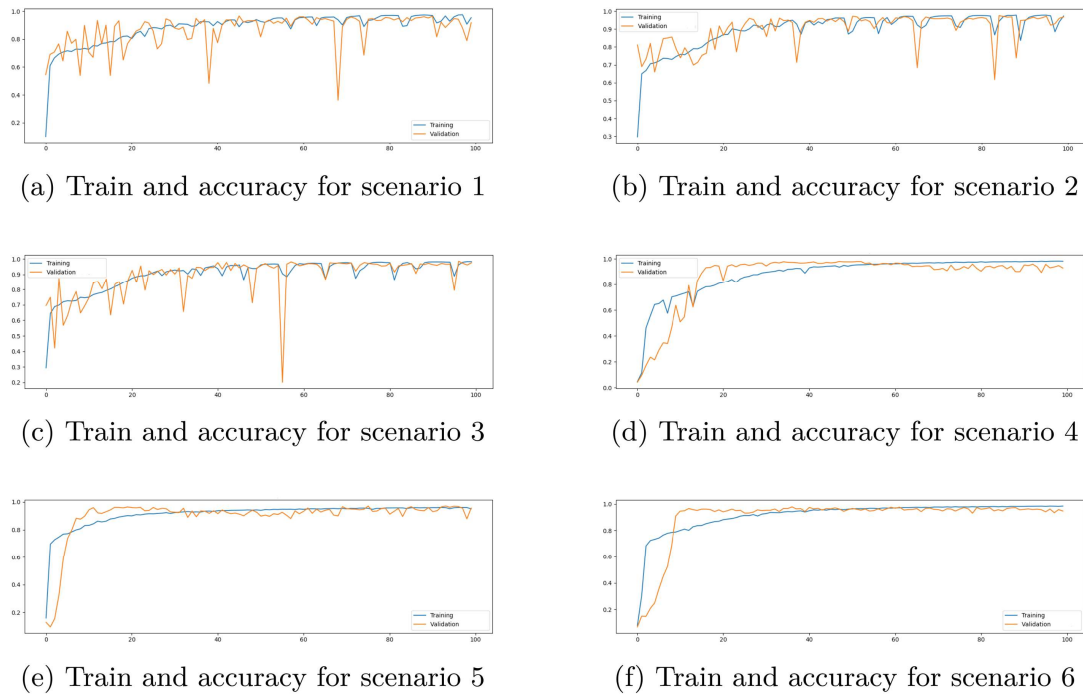


Fig. 3. Figure illustrates a comparison of accuracy achieved across the six scenarios

Table 3. Confidence on the dataset after applying JRIP and PART algorithm

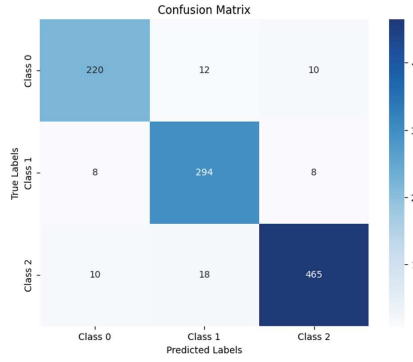
Algorithm	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)
U-Net JRIP	98.66	98.60	98.53	98.57
U-Net PART	96.74	96.14	96.08	96.11
DeepLabv3+ JRIP	94.83	94.09	94.03	94.06
DeepLabv3+ PART	91.96	91.35	91.60	91.46

Table 4. Model Performance Metrics

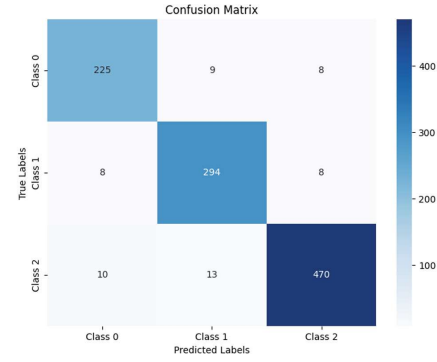
Scenario	Class	Precision(%)	Recall(%)	F1-Score(%)	mAP(%)	IoU
1	Benign	92.44	90.91	91.67	93.68	0.875
	Malignant	90.74	94.84	92.74		
	Normal	96.27	94.32	95.29		
2	Benign	92.59	92.98	92.78	94.64	0.899
	Malignant	93.04	94.84	93.93		
	Normal	96.71	95.33	96.02		
3	Benign	88.31	90.50	89.39	92.25	0.857
	Malignant	89.91	91.94	90.91		
	Normal	95.83	93.31	94.55		
4	Benign	84.85	80.99	82.88	88.61	0.796
	Malignant	85.42	92.58	88.85		
	Normal	92.68	89.86	91.25		
5	Benign	85.71	82.50	84.08	89.66	0.813
	Malignant	87.20	91.67	89.38		
	Normal	93.21	91.89	92.54		
6	Benign	84.98	81.82	83.37	88.90	0.8
	Malignant	85.67	92.58	88.99		
	Normal	93.08	90.06	91.55		

Table 4 presents a comparison of the six experimental scenarios, including the integration of JRIP and PART algorithms, across three categories: benign, malignant, and normal. For Sce. 4–6, the JRIP-based approach delivers the highest accuracy, reaching 89.66% in Sce. 5. This represents an improvement over the baseline model in Sce. 4 (88.61%) and Sce. 6 (88.90%), with notable gains in identifying malignant cases. In contrast, Sce. 1–3 demonstrate stronger overall performance, with Sce. 1 achieving 93.68% accuracy and consistently good results across all classes. The best outcome is obtained in Sce. 2, which achieves 94.64% accuracy, along with balanced precision, recall, and F1-score across the three categories. Consequently, Sce. 2 emerges as the most effective and stable configuration, making it the optimal choice for this classification task.

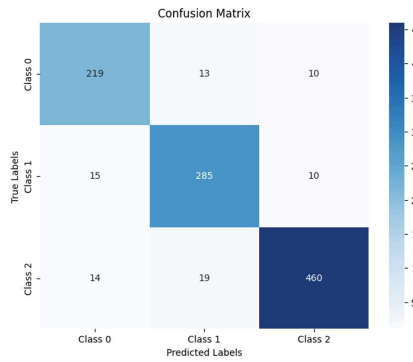
Analysis of the confusion matrices for the six scenarios (with 0: Benign, 1: Malignant, 2: Normal) further illustrates the differences in classification capability. Models built with the U-Net architecture show superior results overall. Class 2 (Normal) consistently records the highest prediction accuracy, as exemplified by Sce. 2, where 470 out of 493 samples are correctly classified. However, frequent misclassification occurs between classes 0 and 1 (Benign vs. Malignant), partic-



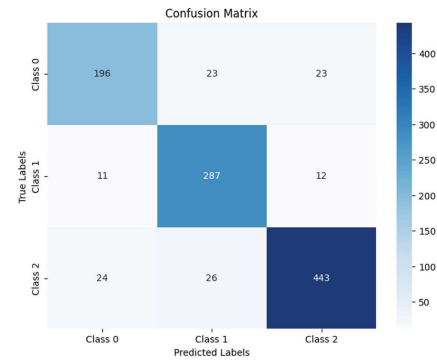
(a) Confusion matrix for Sce. 1



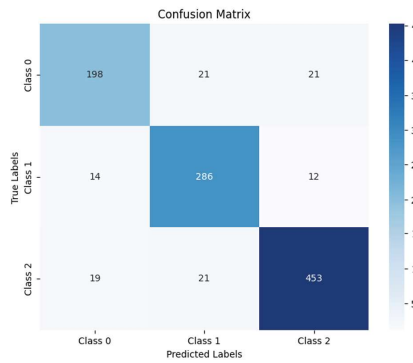
(b) Confusion matrix for Sce. 2



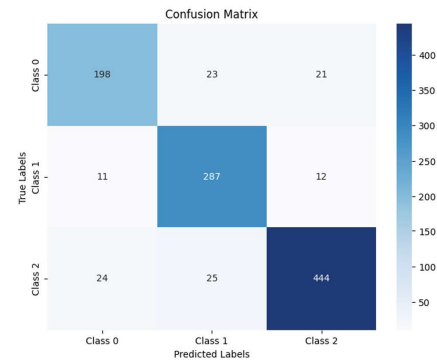
(c) Confusion matrix for Sce. 3



(d) Confusion matrix for Sce. 4



(e) Confusion matrix for Sce. 5



(f) Confusion matrix for Sce. 6

Fig. 4. Confusion matrix of the six scenarios.

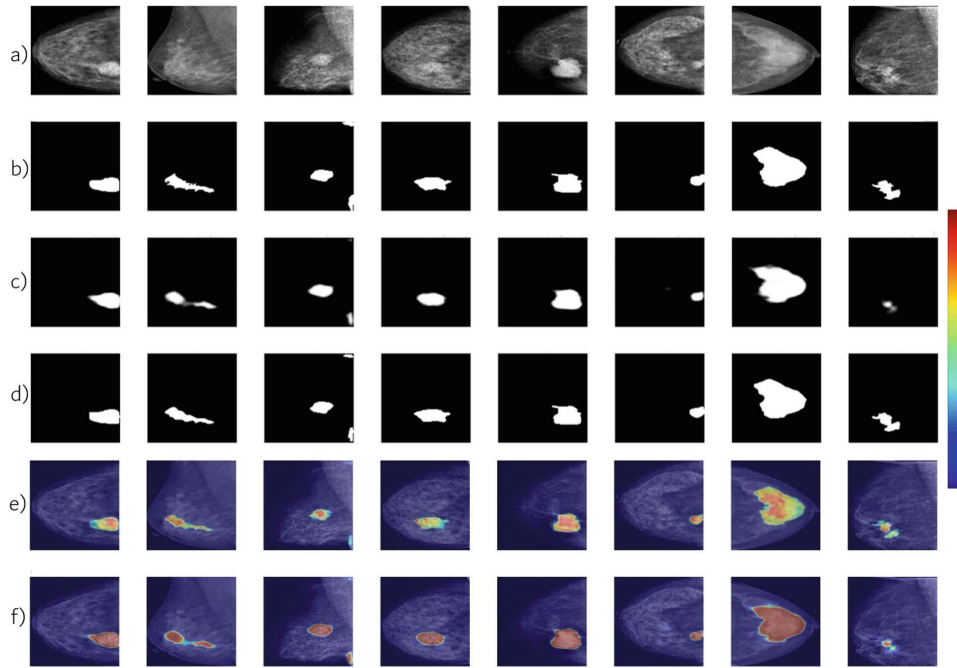


Fig. 5. The classification results on the test set a) Original image, b) Ground truth mask, c) Mask of DeepLabv3+ model, d) Mask of U-Net model, e) U-Net Grad-CAM, f) DeepLabv3+ Grad-CAM

ularly in Sce. 4, 5, and 6. This can be attributed to the similarity of features in these two categories, combined with the smaller number of samples compared to the Normal class, which limits generalization. Sce. 1 and 2 alleviate this issue, showing stronger recognition of class 0, while class 1 (Malignant) also improves substantially in these scenarios. Among all, Sce. 2 delivers the most balanced and reliable performance across the three classes (Fig. 4).

Figure 5 illustrates the experimental results on the mammogram test set for the proposed scenarios. Figure (a) is the original image, and Fig. (b) is the ground truth mask, which is used to mark the mass region. Figure (c) and (d) show the segmentation results obtained by the U-Net and DeepLabv3+ models (scenarios 2, 5), respectively, in which both DeepLabv3+ and U-Net models give good segmentation results when compared with the ground truth mask. Figure (e) and (f) are the Grad-CAM visualizations of these two models, which help to identify which regions in the image have the greatest influence on the model's decision. The red regions in the Grad-CAM show the regions with the highest activation levels corresponding to the mass region, reflecting the model's focus on the abnormal region, which is the region that has an impact on breast cancer identification and classification. The Grad-CAM results can provide more information about how each model pays attention to features in the image. From there, the performance between U-Net and DeepLabv3+ can be compared to choose the most suitable model for the medical image segmentation task.

5 Conclusion

In this paper, we propose a multi-task model for detecting, classifying, and segmenting breast cancer based on rule sets and XAI. By using machine learning to identify important characteristics, we apply rule generation methods to get rid of less important features, which helps the classification model perform better. In addition, we combine the fuzzy inference system with the Grad-Cam technique to display the importance of the detected mass region, thereby improving the efficiency of explaining the output of the proposed method. Experimental results demonstrate that the U-Net_JRIP combination model achieves an accuracy of 94.64%. Also, some combination algorithms haven't performed well because the problem is too general across many layers and the images in the classification layers look very similar, especially between the benign and malignant layers. In future work, we will compare and improve the rule generation models and incorporate additional features such as microcalcifications and density in the images to improve the model accuracy.

References

1. Chong, A., et al.: Digital breast tomosynthesis: concepts and clinical practice. *Radiology* **292**(1), 1–14 (2019)
2. Chowdhary, C.L., et al.: Past, present and future of gene feature selection for breast cancer classification—a survey. *Int. J. Eng. Syst. Model. Simul.* **13**(2), 140–153 (2022)
3. Clancy, K., et al.: Deep learning pre-training strategy for mammogram image classification: an evaluation study. *J. Digit. Imag.* **33**, 1257–1265 (2020)
4. Apoorva, V., Yogish, H.K., Chayadevi, M.L.: Breast cancer prediction using machine learning techniques. In: 3rd International Conference on Integrated Intelligent Computing Communication & Security (ICIIC 2021). Atlantis Press (2021)
5. Prodan, M., Paraschiv, E., Stanciu, A.: Applying deep learning methods for mammography analysis and breast cancer detection. *Appl. Sci.* **13**(7), 4272 (2023)
6. Hui, J.: mAP (mean Average Precision) for object detection. Medium. com. (2018). <https://jonathan-hui.medium.com/map-mean-averageprecisionfor-object-detection-45c121a31173>. Accessed 14 May 2021
7. Cohen, W.W., Singer, Y.: A simple, fast, and effective rule learner. *AAAI/IAAI* **99**(335–342), 3 (1999)
8. Quinlan, J.R.: Induction of decision trees. *Mach. Learn.* **1**, 81–106 (1986)
9. Zadeh, L.A.: A fuzzy-algorithmic approach to the definition of complex or imprecise concepts. *Int. J. Man Mach. Stud.* **8**(3), 249–291 (1976)
10. Ronneberger, O., Fischer, P., Brox, T.: U-Net: convolutional networks for biomedical image segmentation. Medical image computing and computer-assisted intervention—MICCAI 2015: 18th international conference, Munich, Germany, 5–9 October 2015, proceedings, part III 18. Springer international publishing (2015). https://doi.org/10.1007/978-3-319-24574-4_28

11. Chen, L.-C., et al.: Encoder-decoder with Atrous separable convolution for semantic image segmentation. In: Proceedings of the European Conference on Computer Vision (ECCV) (2018)
12. Lizzi, F.: Development and discussion of deep learning algorithms for breast density classification and for COVID-19 lesions quantification on CT scans: towards a real-time multidisciplinary approach (2022)